



DO-IT-YOURSELF: PROJECT

The variables GreenState, YellowState record the states of the transistors connected to green and yellow LEDs. Since the output is 180-degree phase shifted, 'HIGH' corresponds to a transistor being off and 'LOW' corresponds to the transistor being on. Fig. 3, Fig. 4, and Fig. 5 show snapshots of the source code.

The data from this variable is further used to define if else logic to control the solenoid and the water pump. The LCD used to display the state of the two water sources is controlled using I2C communication. The libraries, Wire.h and LiquidCrystal_I2C.h, are utilised for I2C communication with the LCD. After each loop, a delay of 2000ms is added in the execution of the code.

You can download the complete code from electronicsforu.com. After getting the code ready, upload the code to Arduino by selecting the right port and board.

Construction and testing

You can assemble the circuit on a small general-purpose PCB. But before assembly do not forget to upload the source code into Arduino Nano. The project has two parts, the mechanical assembly and the electrical circuit. The pipe mechanical assembly is shown in Fig. 6.

The mechanical part of this project can be constructed using 1.3cm (½-inch) dia PVC pipe and corresponding fittings. The following fittings may be required:

- Elbow connector
- 'T' connector
- Female copper adaptor
- Steel hose nipple

Two metal nails are inserted into the PCV pipe to act as electrode to detect the availability of municipal supply, as mentioned earlier.

The complete circuit is soldered on a perf board as per the circuit diagram shown in Fig. 2. Terminal blocks are utilised to connect the



Inside View



Top View

Fig. 8: Inside and top views of the circuit in an enclosure



Fig. 9: Side views showing different connectors



Fig. 10: Complete pipe assembly

12V DC power, level probe, electrode, and solenoid. The three LEDs are soldered on a separate perf board and mounted on the enclosure. Heat-sinks are used for the MOSFETs to dissipate heat generated during their operation.

The complete assembly is shown in Fig. 8, Fig. 9, and Fig. 10.

The circuit is enclosed inside a plastic enclosure with the lid used to

mount the LCD and the LEDs. Along with the circuit board, the 12V, 3.3A SMPS and the contactor are also fitted inside the enclosure.

On the sides of the enclosure, different connectors are mounted to connect the sensing elements and devices, as shown in Fig. 9. These connectors are:

- JST connectors to connect the water level probe and the electrodes